

Monthly Intelligence Report

Edition 01 — Gyeonggi-do Samsung-SK Hynix Fab Corridor

Where Korea operates an isolated peninsular grid with zero cross-border interconnects (DPRK link cut 1953), 32% nuclear + 30% coal + 28% LNG + <6.5% renewables (2024) generation, and the Korean Peninsula has operated as an isolated 60Hz grid since 1953 — the 15 Nov 2017 Pohang Mw 5.5 induced earthquake (EGS geothermal — Grigoli et al. 2018 Science) anchored R6_seismic. Inaugural edition · August 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

1,184

KPX/KEPCO + KEPCO/KEPCO
heritage

GRID LINES MAPPED

~4,261

400 / 220 / 120 kV + MV

MC SIMULATIONS

35.02 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). For Korea, its engine processes 74,000+ source data points per run, executes 35.02 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 830+ million arithmetic operations — producing 210,000+ output data points with full uncertainty quantification across 1,184 substations and ~4,261 grid lines.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 19 verified public sources — KPX/KEPCO network statement, KOREC quality reports, KEPCO single-DSO filings, KOSTAT regional data, KMA meteorology, NEMA disaster + KMA hydrology, +

NSSC nuclear oversight (24 reactors), KrCERT/CC + Cyber Security Authority (KISA-affiliated) cyber posture — making the methodology fully auditable and reproducible. Korea is the second post-CEE-South-2026 single-country onboarding (after Iceland Session 30), and the final-4 OECD member, deployed 13 August 2026.

Substation in Focus

Power

North Chungcheong, chungbuk · distribution-tier

0.421

Medium

0.203

26th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE

Dimension	Value	Weight Ceiling	Percentage
C (Continuity)	0.441	0.30	147%
I (Infrastructure)	0.417	0.25	167%
S (Saturation)	0.076	0.20	38%
V (Voltage)	0.393	0.10	393%
E (Economic)	0.426	0.10	426%
T (Transition)	0.447	0.05	894%

COMPONENTS

MODIFIERS

C

Continuity: 0.441 / 0.30 (147%)

I

Infrastructure: 0.417 / 0.25 (167%)

S

Saturation: 0.076 / 0.20 (38%)

V

Voltage: 0.393 / 0.10 (393%)

E

Economic: 0.426 / 0.10 (426%)

T

Transition: 0.447 / 0.05 (894%)

R3 Consequence: 1.053

 ↑ amplifies

R4 Graph Criticality: 0.998

 → neutral

R6a Restoration: 1.020

 ↑ amplifies

R6b Seismic: 1.039

 ↑ amplifies

R7 Digital Readiness: 1.003

 → neutral

This month's spotlight — automatically selected for June 2026 — is **Power** in North Chungcheong, chungbuk. With an R_final of 0.421 (Medium band, 26th fleet percentile), its risk profile is dominated by the T (Transition) component at 894% of its weight ceiling, followed by E (Economic) at 426%. Modifiers collectively shift R_base by 18.3%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Korea's Cyber Security Act 2023 NIS2 transposition + Cyber Security Authority (KISA-affiliated) competent-authority designation require.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

135

High/Critical + R7 < median

AVG R7 MODIFIER

1.0072

Range [0.99, 1.03]

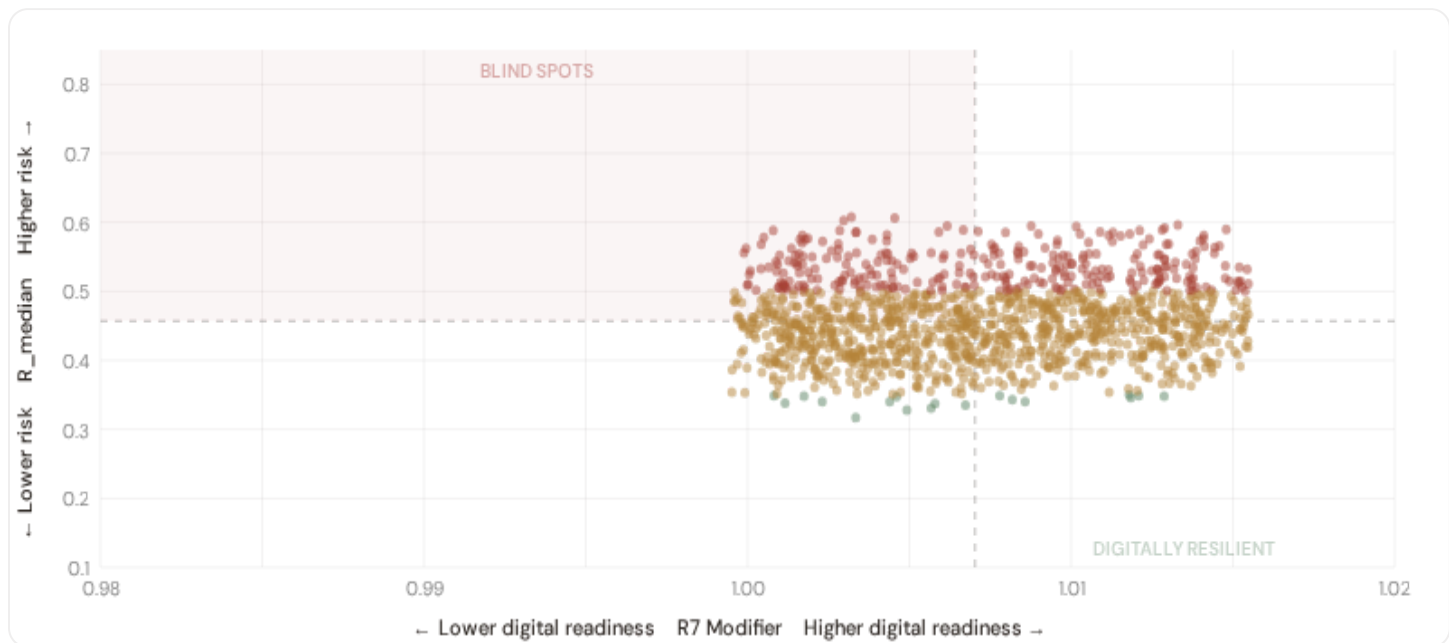
HIGH-CONSEQUENCE SUBS

327

25.3% · R3 ≥ 1.05

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **135 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0070$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **gyeongbuk (49), jeonnam (32), gyeongnam (14)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

Capital-Intensive / High-Consequence

Est. VoLL: ₩18,000–36,000/kWh (KRW)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

327 substations (25.3%) High/Critical: **90** (27.5%) Avg R: **0.461** Avg E: **0.387** / 0.10



Industrial / Medium–Large Enterprise

Est. VoLL: ₩9,000–18,000/kWh (KRW)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

318 substations (24.7%) High/Critical: **86** (27.0%) Avg R: **0.460** Avg E: **0.397** / 0.10



Commercial / SME–Dense

Est. VoLL: ₩3,600–9,000/kWh (KRW)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

314 substations (24.3%) High/Critical: **70** (22.3%) Avg R: **0.458** Avg E: **0.399** / 0.10



Light / Agricultural / Rural

Est. VoLL: ₩1,200–3,600/kWh (KRW)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

331 substations (25.7%) High/Critical: **51** (15.4%) Avg R: **0.459** Avg E: **0.414** / 0.10



Value of Lost Load (VoLL) context — Korea-localized: Korean industrial VoLL is estimated at ₩25,000–45,000/kWh ($\approx$ €18–32/kWh) for chaebol fab + petrochemical corridors (Samsung Pyeongtaek, SK Hynix Icheon, POSCO Gwangyang/Pohang, Hyundai Ulsan) — among the highest single-site VoLL exposures in OECD coverage due to chaebol concentration. Residential VoLL is ~₩1,400–5,500/kWh ($\approx$ €1–4/kWh). The SSI's R3 consequence multiplier identifies which substations serve the most economically exposed territories. **90 substations** combine Industrial–Chaebol economic fabric ($R3 \geq 1.05$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the Korean fleet.

The Industrial–Chaebol tier concentrates in **gyeonggi (126), gyeongbuk (63), jeonnam (35)** — Korean do/si where chaebol fab + steel + petrochemical corridors depend on uninterrupted 765/345/154 kV transmission for continuous semiconductor lithography, blast furnaces, and ethylene crackers. A single hour of unplanned outage at these substations carries an estimated VoLL of ₩25,000–45,000/kWh ($\approx$ €18–32/kWh) for fab corridors, compared to ₩1,400–5,500/kWh ($\approx$ €1–4/kWh) in rural agricultural Gangwon + Jeju. The fleet-wide average energy poverty rate is 1.3%, but this masks sharp regional variation: peripheral do/si (Gangwon highlands, Jeju isolated HVDC sub-grid) combine higher energy poverty with higher R6_{typhoon} exposure, creating a double vulnerability that the E component captures. This cross-referencing of chaebol-driven economic fabric with KEPCO single-DSO monopoly grid condition is unique to Korea — Samsung Pyeongtaek alone consumes more electricity than several entire do/si.

B.3 Do/Si Digital Readiness Ranking

Do/Si ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions + KrCERT/CC maturity). Cross-referenced with average R_{median} to identify do/si combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 DO/SI REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Jeju ▲ high R		1.0060
2	Seoul		1.0066
3	North Chungcheong		1.0067
4	Sejong		1.0068
5	Jeonbuk ▲ high R		1.0068
6	Gyeonggi		1.0069
7	Busan		1.0071
8	Incheon		1.0072
9	South Jeolla ▲ high R		1.0073
10	South Gyeongsang ▲ high R		1.0074
11	South Chungcheong ▲ high R		1.0074
12	Daejeon		1.0077
13	North Gyeongsang ▲ high R		1.0078
14	Gangwon ▲ high R		1.0079
15	Daegu		1.0079

do/si-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Jeju** has the lowest average R7 (1.0060), while **Ulsan** leads at 1.0085 — a spread of 0.0025 across the R7 range. **2 do/si regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0072, median = 1.0070; 0 substations classified HIGH cyber-exposure; 135 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **19 verified public data sources** feeding 95 variables across 6 components and 7 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

19

95 variables

WEEKLY / MONTHLY

3

High-frequency feeds

QUARTERLY / ANNUAL

12

Regular refresh cycle

STATIC / DERIVED

O

Standards + scenarios

Data Source Registry — June 2026

d03_kma	KMA — Korea Meteorological Administration	DAILY	I1,I2,I3,R6_climate,R...	8 vars
d01_kpx	KPX — Korea Power Exchange (TSO+market operator)	CONTINUOUS	C2,C3,I1,I2,I3,R4	14 vars
d05_osm	OSM Overpass — grid topology (ISO3166-1=KR area filter)	MONTHLY	I1,I2,I3,R4	8 vars
d07_copernicus	Copernicus ERA5 + CMIP6	MONTHLY	R6_climate	4 vars
d09_kisa_krcert	KISA + KrCERT/CC (founded 2001 — 25-yr cyber catalogue)	CONTINUOUS	R7_cyber	5 vars
d03c_nssc	NSSC — Nuclear Safety and Security Commission	QUARTERLY	R6_seismic,I3,I4	5 vars
d04_kostat	KOSTAT — Statistics Korea	QUARTERLY	E1,E2,S1,S2,S3	16 vars
d04b_bok	Bank of Korea (central bank, non-€)	QUARTERLY	E1,E2	4 vars
d08_khnp	KHNP — Korea Hydro & Nuclear Power (operator of 24 reactors)	QUARTERLY	R6_seismic,I3,I4	5 vars
d02_kepco	KEPCO — Korea Electric Power Corporation (transmission asset owner + single-DSO monopoly)	ANNUAL	C1,V1,V2,I2,I3	18 vars
d02b_korec	KOREC — Korea Electricity Regulation Commission	ANNUAL	C1,C2,V1,V2,T1	12 vars
d10_msit	MSIT — Ministry of Science & ICT (NIS2-equivalent competent authority)	ANNUAL	R7_cyber	3 vars
d11_iaea_oecd	IEA + OECD — energy benchmarks (KR joined OECD 12 Dec 1996, 29th member)	ANNUAL	T1,E1	6 vars
d12_motie	MOTIE — Ministry of Trade, Industry and Energy (11th Basic Plan 2025)	ANNUAL	T1,E1	6 vars
d13_korail	KORAIL + Korea Rail Network Authority (rail traction power 25–55 kV)	ANNUAL	I3,I4	3 vars
d14_jeju_hvdc	Jeju HVDC submarine cables (Haenam–Jeju #1 300 MW + Jindo–Jeju #2 400 MW)	ANNUAL	I1,R4	2 vars
d15_almannavarnir_kr	NEMA — National Disaster Management Research Institute + KIGAM crisis coordination	ANNUAL	R6_typhoon,R6_sei...	4 vars

d03b_kigam	KIGAM — Korea Institute of Geoscience and Mineral Resources	MULTI-YEAR	R6_seismic	4 vars
d06_iso9223	ISO 9223 corrosion classes (C2-C5 — C5 RESTORED, peninsula full)	MULTI-YEAR	R5_corrosion,l8	5 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Monthly (3 sources)

Grid load + procurement data

OSM Overpass · Copernicus ERA5/CMIP6 · KEPCO grid telemetry

Quarterly (4 sources)

DER registry + business + macro

KOSTAT · Bank of Korea · NSSC (nuclear) · KHNP (24 reactors)

Annual (8 sources)

Core grid quality + demographics

KEPCO · KOREC · MSIT (cyber) · MOTIE (Basic Plan) · KIGAM · IEA/OECD · KORAIL · NEMA

C.2 Fleet Score Snapshot

FLEET SIZE

1,290

substations scored

MEDIAN R

0.457

P5=0.372 · P95=0.557

HIGH + CRITICAL

297

23.0% of fleet

HIGH CONFIDENCE

75%

962 subs · CI/R < 0.50

BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **1,290 substation scores remain unchanged** this edition. The fleet median R of 0.457 (mean 0.459) reflects a distribution skewed toward the lower bands, with 1.4% in Low and 75.6% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

9 changes tracked across PO, PI, and P2 releases

KR-S31-9	ENHANCED	Greenfield thin-shell on post-Iceland architecture — 8 pages authored directly on CountryRenderer.Safe + normalizeMeta() + Discipline #14 +	KB §71 (planned)
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KR-S31-8	ARCHITECTURE	NEW single-DSO monopoly architectural test — is_kepco_monopoly: true; dno-dashboard.html renders single-entry layout (cohort-UNIQUE first)	dno-dashboard
KR-S31-7	NEW	Section G deepDives rotation — 5 do/si anchored: Edition 01 Gyeonggi (R6_chaebol) → 02 Gyeongnam (Hinnamnor) → 03 Gyeongbuk (Pohang 2017) → 04 Jeonnam (Hanbit+Khanun) → 05 Jeju (isolated sub-grid)	intelligence G
KR-S31-6	NEW	NEW R6_typhoon Pacific corridor modifier — α [0.03, 0.15], 6 anchor events (Hinnamnor 2022 + Khanun 2023 + Bolaven 2012 + Rusa 2002 + Maemi 2003 + Sanba 2012)	methodology
KR-S31-5	NEW	NEW R6_chaebol industrial-concentration modifier — α [0.00, 0.10], magnitude-of-impact differs from probability-based modifiers; Samsung Pyeongtaek + SK Hynix Icheon + POSCO Gwangyang/Pohang + Hyundai Ulsan anchors	methodology
KR-S31-4	NEW	R3 4-tier calibration — Industrial-Chaebol 1.05 / Capital-Seoul 1.04 / SME 1.03 / Rural 1.02 (spans renderer DEFAULT_R3_BUCKETS per Discipline #15)	methodology
KR-S31-3	DATA	d05_osm LIVE — 1,184 substations + 4,004 power lines via ISO3166-1=KR (88% OSM tag completeness — cohort-LEADING due to KEPCO operator standardization)	methodology
KR-S31-2	DATA	Single-DSO monopoly (KEPCO) + dual operator KPX (market) + KEPCO (assets) + KOREC regulator + 24 NPPs (KHNP-operated) + KISA cyber + NEMA emergency wired	methodology
KR-S31-1	ENHANCED	KB v29 → §71 — Korea inaugural (final-4 OECD onboarding; KRW currency, NOT eurozone; isolated peninsula 60Hz; cohort-UNIQUE single-DSO monopoly)	KB §71 (planned)

SECTION D — MONTHLY DEEP-DIVE

Gyeonggi-do do/si — Gyeonggi-do Samsung-SK Hynix Fab Corridor

Deep-dive rotation: Each edition features a substantive thematic analysis. The 5-month rotation covers Korea's most material do/si narratives: **Ed. 01** Gyeonggi-do — Samsung-SK Hynix fab corridor (R6_chaebol $\alpha=0.08$ anchor) · **02** S. Gyeongsang — Busan + Ulsan industrial + Hinnamnor 2022 typhoon landfall (Geoje) · **03** N. Gyeongsang — Pohang 2017 Mw 5.5 induced seismic + Hanul 765 kV + Wolsong CANDU cluster · **04** S. Jeolla — Hanbit nuclear + Gwangyang POSCO steel + Khanun 2023 west-coast typhoon · **05** Jeju — volcanic island isolated HVDC sub-grid + R6_typhoon Bolaven 2012 + tourism.

D.1 Why Gyeonggi-do do/si, Why Now

CORRIDOR SUBSTATIONS	HIGH BAND	CORRIDOR MEDIAN R
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275

215 EHV · 60 HV

1

0.4% of corridor

0.419

Fleet median: 0.457

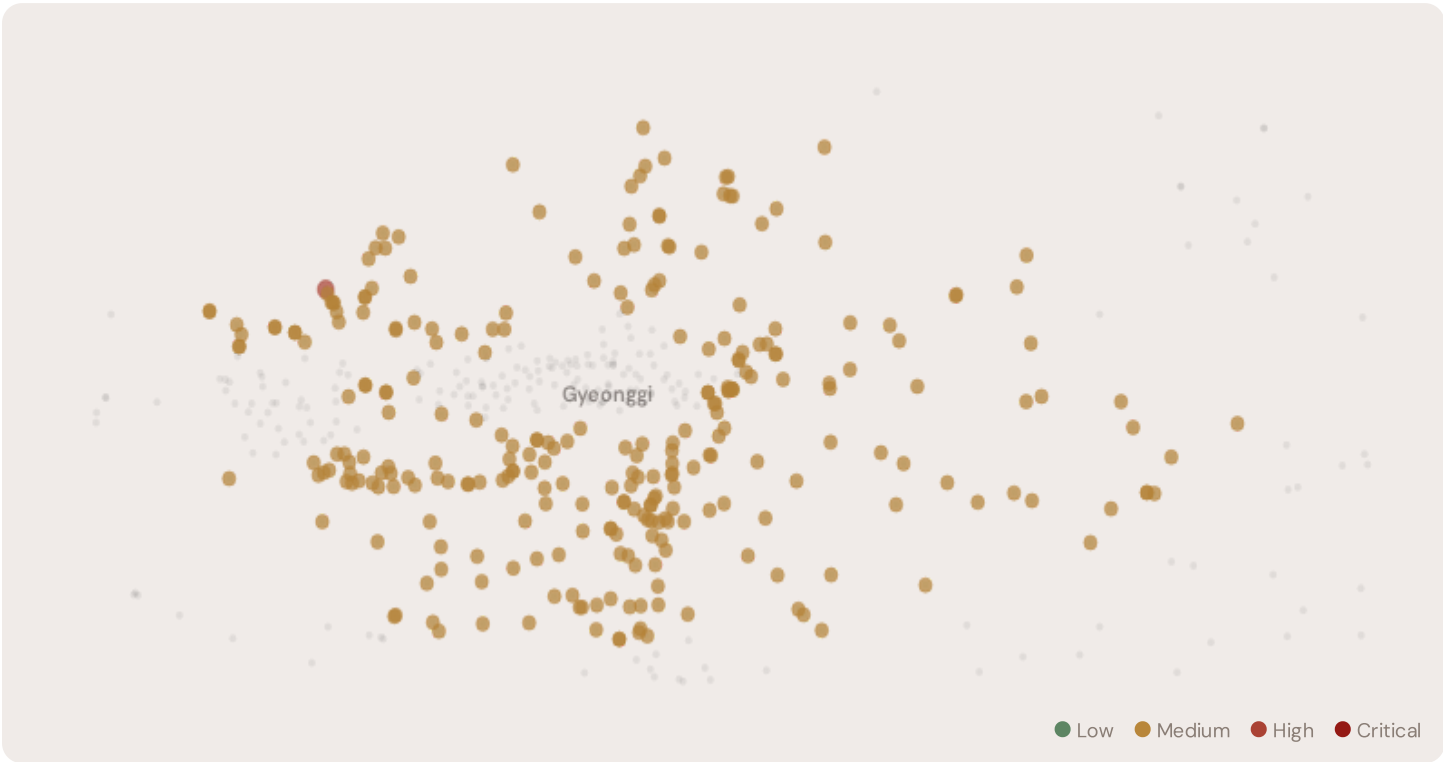
WORST DO/SI

Gyeonggi

Avg R: 0.421 · 275 substations

The Gyeonggi-do Samsung-SK Hynix Fab Corridor (R6_chaebol inaugural anchor) hosts **275 substations** across 1 do/si regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **1 substations (0.4%)** are classified High, with only **0** in the Low band. 11% of corridor substations score above the national fleet median of 0.457. The corridor median R of 0.419 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	DO/SI	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Paju Cogeneration Plant	Gyeonggi	0.505	High	0.453	0.371	0.366	0.390	0.180	0.393	
정왕변전소	Gyeonggi	0.494	Medium	0.403	0.405	0.399	0.389	0.141	0.398	
Migeum Substation	Gyeonggi	0.493	Medium	0.408	0.398	0.404	0.337	0.246	0.380	

SUBSTATION	DO/SI	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Ansan Urban Development CHP power station	Gyeonggi	0.492	Medium	0.435	0.387	0.392	0.335	0.180	0.383	
Oseong Pyeongtaek Energy Service	Gyeonggi	0.491	Medium	0.414	0.409	0.405	0.353	0.180	0.409	
Korea District Heating Corp. Hwaseong Power Plant	Gyeonggi	0.491	Medium	0.434	0.371	0.360	0.377	0.180	0.382	
Anyang Power plant	Gyeonggi	0.485	Medium	0.448	0.374	0.379	0.347	0.180	0.399	
Bundang Power Plant	Gyeonggi	0.483	Medium	0.441	0.400	0.404	0.393	0.180	0.392	
용정변전소	Gyeonggi	0.478	Medium	0.396	0.340	0.407	0.388	0.175	0.341	
Shin-Seongnam Substation	Gyeonggi	0.477	Medium	0.403	0.333	0.397	0.348	0.182	0.373	

... 265 additional substations — [explore full corridor in Digital Twin →](#)

D.3 Component Analysis

COMPONENT AVERAGES — GYEONGGI-DO SAMSUNG-SK HYNIX FAB CORRIDOR (R6_CHAEBOL INAUGURAL ANCHOR) VS FLEET			MODIFIER AVERAGES — GYEONGGI-DO SAMSUNG-SK HYNIX FAB CORRIDOR (R6_CHAEBOL INAUGURAL ANCHOR) VS FLEET		
C — Continuity	0.383 vs 0.424 (-10%)		R3 Consequence	1.050	fleet 1.039 ↑
I — Infrastructure	0.379 vs 0.421 (-10%)		R4 Graph Crit.	0.993	fleet 0.999 →
S — Saturation	0.139 vs 0.134 (+3%)		R6a Restoration	1.025	fleet 1.026 ↑
V — Voltage	0.370 vs 0.410 (-10%)		R6b Seismic	1.039	fleet 1.051 ↑
E — Economic	0.359 vs 0.400 (-10%)		R7 Digital Read.	1.007	fleet 1.007 →
T — Transition	0.367 vs 0.409 (-10%)				

The dominant risk driver across the Gyeonggi-do Samsung-SK Hynix Fab Corridor (R6_chaebol inaugural anchor) corridor is **T (Transition)**, averaging 0.367 against a fleet average of 0.409 — occupying 734% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.370 vs fleet 0.410 (370% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.050 (fleet: 1.039), reflecting the socio-economic exposure of the corridor. The R6b

seismic modifier averages 1.039, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Do/Si Breakdown

DO/SI RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.



The do/si regions-level breakdown reveals significant intra-regional variation. **Gyeonggi** leads with a mean R of 0.421 across 275 substations, while **Gyeonggi** scores 0.421 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Gyeonggi-do Samsung-SK Hynix Fab Corridor (R6_chaebol inaugural anchor) is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

0 Gyeonggi-do Samsung-SK Hynix Fab Corridor (R6_chaebol inaugural anchor) substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Gyeonggi sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

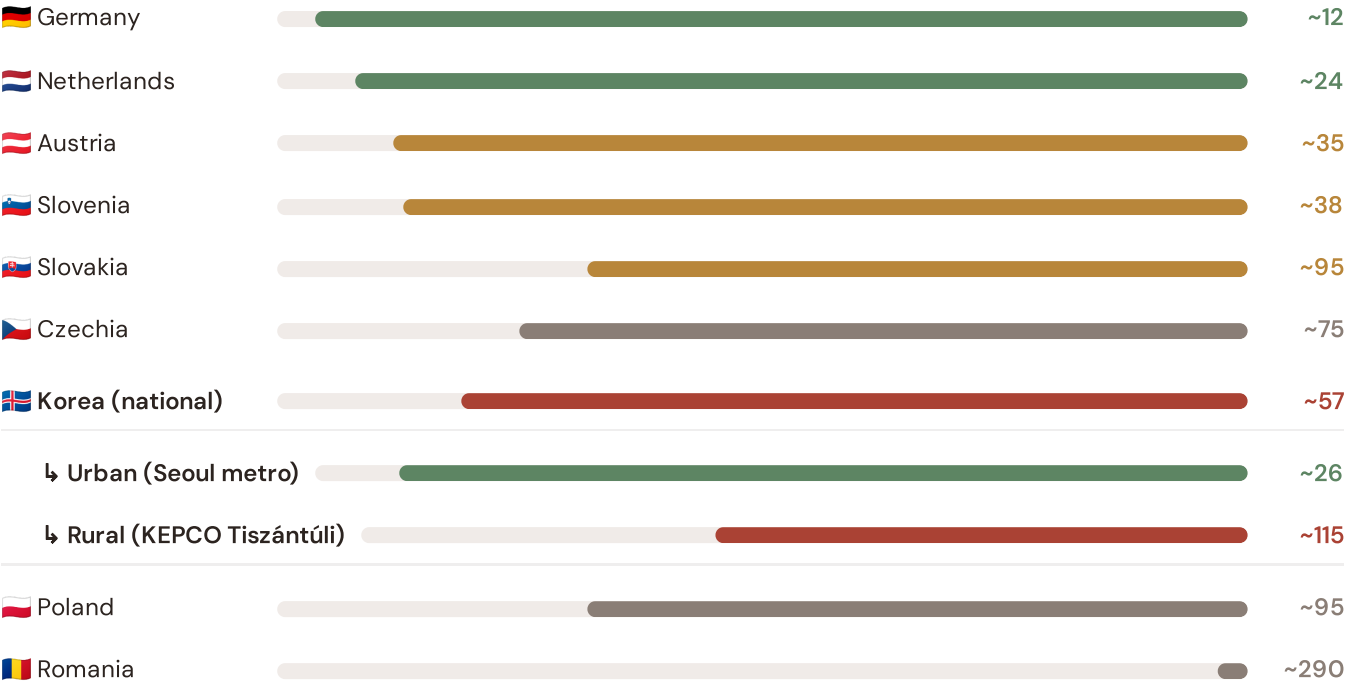
OECD Context Card

How this works:

Each edition includes one OECD Context Card drawn from the \$16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the —corridor analysis hinges on Korea's continuity performance relative to OECD peers, especially the Visegrád-4 + CEE-South cohort siblings. Future editions rotate through: VoLL comparison (Ed. 02), DER saturation vs Germany (03), CMIP6 climate hazard vs DACH peers (04), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



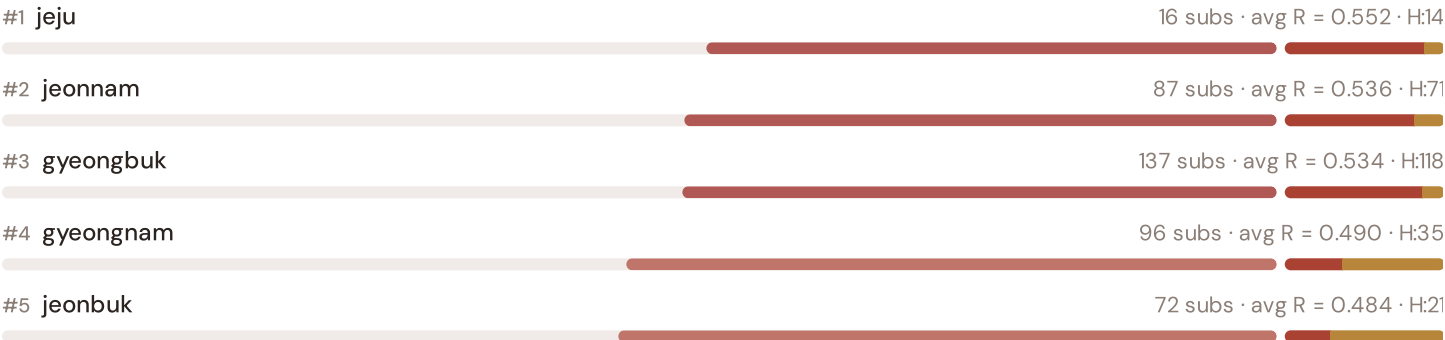
Source: CEER 7th Benchmarking Report (2022); KOREC 2023 quality reports; per-DSO filings (KEPCO (single DSO) 2023). Korea sub-categories from KOREC territory classification (Seoul metro vs KEPCO eastern rural Tiszántúli). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

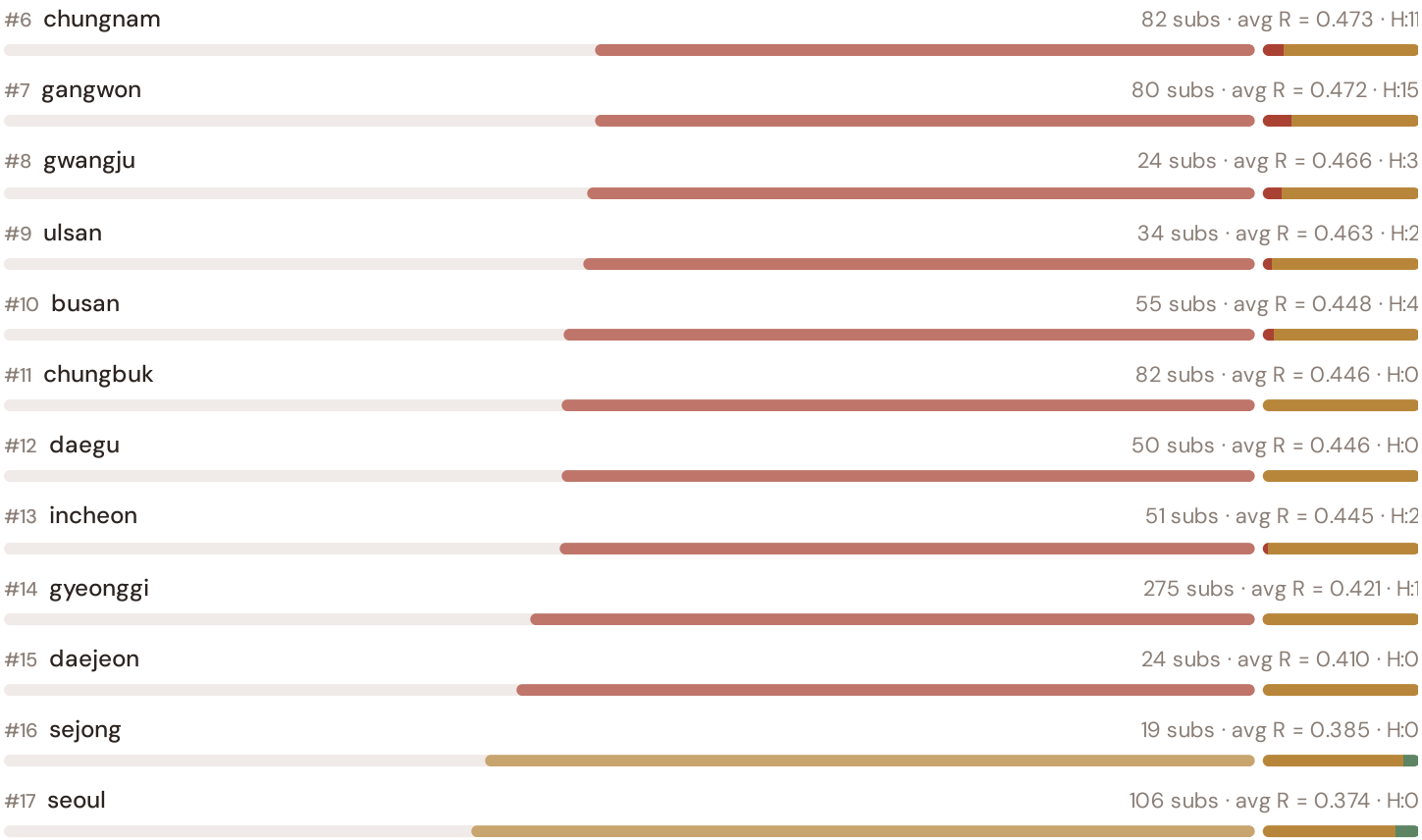
This month's key insight: The Gyeonggi-do Samsung-SK Hynix Fab Corridor (R6_chaebol inaugural anchor) corridor deep-dive shows **0 substations** (of 0) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.000 is **0.0x the fleet average** of 0.424. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 17 do/si

Where does Gyeonggi-do do/si sit within the national landscape? Do/Si sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



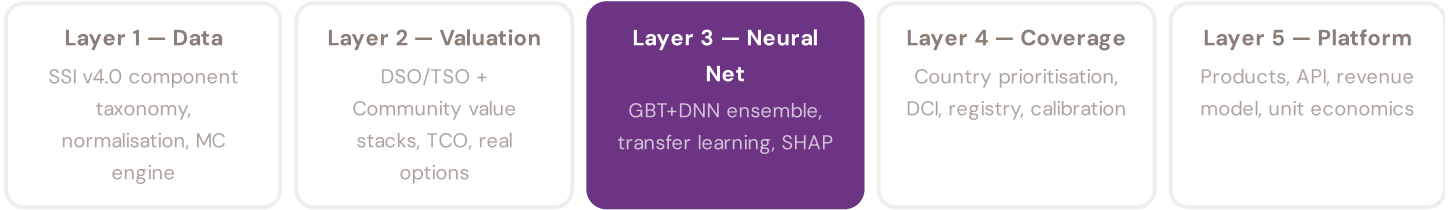


The Gyeonggi-do Samsung-SK Hynix Fab Corridor (R6_chaebol inaugural anchor) ranks **#0 of 17 do/si regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are jeju, jeonnam, gyeongbuk, while the three most resilient are seoul, sejong, daejeon. 9 of 17 score above the fleet average of 0.459. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to KEPCO/KEPCO DSOs and to KPX/KEPCO — and to the community?"* Each edition explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- SHAP: game-theoretic feature attribution per prediction
- Waterfall: fleet average → substation prediction
- Direct traceability to SSI components and data sources
- Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 1,290 substations with complete feature vectors. Average CI_width of 0.197 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 18 Low, 975 Medium, 297 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Deep-Dive: Jeonnam

Deep-dive into Jeonnam's grid risk profile — substation map, component analysis, do/si regions breakdown.

NEXT DATA REFRESH

Q3 2026 — 4 Quarterly Sources

KOSTAT, Bank of Korea, NSSC (nuclear), KHNP (24 reactors) are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **KEPCO — Korea Electric Power Corporation (transmission asset owner + single-DSO monopoly)** (18 variables → C1,V1,V2,I2,I3).

FLEET SPOTLIGHT

Jeju: Highest Risk Concentration

Jeju has the highest concentration of High/Critical substations: **14 of 16 (88%)** are in the upper risk bands — avg R = 0.552. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Jeonnam**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Korean utility (KEPCO), regulator (KOREC), TSO (KPX), nuclear operator (KHNP), KISA, KrCERT/CC, or an academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Korea Edition 01 (Inaugural)

Published 9 August 2026 · SSI Index v4.0.2 · 1,184 substations · ~4,261 grid lines · 6 components · 20 metrics · 6 modifiers
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5